



Topaz

△ Well-formed topaz crystal

Once, it was believed that all yellow gems were topaz, and that all topaz was yellow – however, neither statement is true. Some topaz is yellow, but it can also be colourless, blue, green, sherry, and its most valuable colour variation, pink. It is fairly refractive, splitting light into its constituent colours; as a result, colourless topaz resembles diamond, and has often been mistaken for it. Additionally, some blue topaz is almost indistinguishable from aquamarine. A certain number of gems on the market have been treated by heat and irradiation to change their colour.

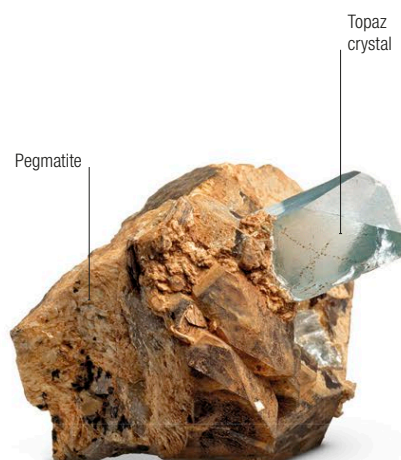
Specification

Chemical name Aluminium fluorosilicate | **Formula** $\text{Al}_2\text{SiO}_4(\text{F},\text{OH})_2$
Colours Yellow, golden, orange, pink, green, blue, colourless
Structure Orthorhombic | **Hardness** 8 | **SG** 3.5–3.6 | **RI** 1.62–1.63 | **Lustre** Vitreous | **Streak** White | **Locations** Brazil, Russia, Germany, Nigeria, Afghanistan, USA, Pakistan, Japan

Rough



Rough crystals | This group of gem-quality topaz crystals shows a clear colour gradation from yellow and sherry to a dark, almost red hue. The outer texture of these crystals obscures their transparent interiors.



Topaz in pegmatite matrix | This finely crystallized, light blue topaz is resting on a groundmass of pegmatite, a mineral host in which it frequently occurs.



Brazilian topaz | The rich, red-brown colour and excellent transparency of this superb topaz crystal from Brazil make it a gem-cutter's dream.

Topaz as a megagem

Gemstone heavyweights

Topaz is found in well formed, prismatic crystals with a lozenge-shaped cross-section. Although most gemstone material is found in alluvial deposits as water-worn pebbles, a number of very large crystals have also been found in situ. The world's largest preserved topaz crystal weighs 271kg (596lb), and in the 1980s a faceted gem weighing 22,892.5 carats – 4.6kg (10.1lb) – was cut from a Brazilian rock fragment.

Sherry topaz crystal | This prismatic crystal shows good form and colour. It comes from the Ouro Preto deposit in Minas Gerais, Brazil, the largest commercial source of sherry topaz.



Brown topaz | A fine, gem-quality topaz, this rough has an unusual brown cast to its colour. Its lightness and transparent surface reveal internal blemishes.